Q) What is switch-case? Explain with example.

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Ans) When one of the many alternatives is to be selected in a C program, we can use an if statement to control the selection. However, the complexity of such a program increases very much when the number of alternatives increases. The program becomes difficult to read & follow. C has a built-in multiway decision statement that allows us to make a decision from the number of choices. This control statement is known as switch.

The switch statement tests the value of a given variable (or expression) against a list of case values and when a match is found, a block of statements associated with that case is executed.

The general form of the switch statement is:-

```
switch (integer expression or characters)
{
        case value-1:
                        block-1
                        break;
        case value-2:
                        block-2
                        break;

        - - -
        - - -
        - - -
        default:        default-block
                        break;
}
```

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Here, value-1, value-2 are constants or constant expressions and are known as case-labels. Each of these values should be unique within a switch statement.

The break statement at the end of each block signals the end of a particular case and causes an exit from the switch statement, transferring the control to the statement following the switch.

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Example of switch-case :-

```
main()
{
    int a=3;
    switch(a)
    {
        case 1:
            printf("\n We are in case 1");
            break;
        case 2:
            printf("\n We are in case 2");
            break;
        case 3:
            printf("\n We are in case 3");
            break;
        default:
            printf("\n We are in default");

    }
}
```

Output will be :-

We are in case 3

Q) Disadvantages of switch statement over if-else :-

1) A float expression cannot be tested using a switch
2) Cases can never have variable expressions (for example, case a+4 is wrong)
3) Multiple cases cannot use same expressions. For example, the following code is wrong :-

```
switch(a)
{
    case 2:
    ----
    case 1+1:
    ----
}
```

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In case of if-else statement, there are no such limitations.

If the conditions in the if-else are simple and less in number, then if-else would work out faster than the lookup mechanism of a switch. Hence, a switch with 2 cases would work slower than an equivalent if-else.

Q) Advantages of switch-case over if-else :-

In a C program, when only one of the many alternatives is to be selected, then the complexity of the program increases very much if we are using if-else ladder. The program becomes difficult to read and follow. switch statement is a much better alternative in such a case.

switch works much faster than an equivalent if-else ladder. This is because the compiler generates a jump table for a switch during compilation. As a result, during execution it simply refers the jump table to decide which case should be executed, rather than actually checking which case is satisfied. As against this, if-else are slower because they are evaluated at execution time. A lookup in the jump table is faster than evaluation of a condition, especially if the condition is complex. Thus, a switch with 10 cases would work much faster than an equivalent if-else ladder.

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